Chapter 4

Forms of corrosion

FORMS OF CORROSSION

Several forms of corrosion can be identified based on the appearance of the corroded metal.

A. Uniform Corrosion.
- Atmospheric Corrosion
- Galvanic Corrosion
- Microbiological Corrosion
- Stray-Current Corrosion

B. Localized Corrosion.
- Pitting Corrosion.
- Stress Cracking Corrosion.
- Erosion-corrosion
- Crevice Corrosion.

A- Uniform Corrosion:

- Uniform corrosion is defined as a corrosion process with a uniform metal dissolution rate over the entire surface area (or a large fraction of the surface area) exposed to the corrosive environment.

- In such type of corrosion there is a uniform decrease in the volume of the metal due to direct contact with the surrounding environment.

- Examples:

1. Dissolution of (Zn and Al metal in acids) and (Pb metal in base)
2. Rust! steel exposed to environment.
   - Uniform in nature
   - leaves scale or deposit over entire exposed area
   - Rust is a hydrated iron-oxide – $Fe(OH)_3$ or $Fe_2O_3.3H_2O$.

Uniform Corrosion:

Atmospheric Corrosion

Atmospheric corrosion takes place due to the direct contact between a metal and its surrounding atmosphere

- Types of atmosphere are encountered:
- Gas type: contain mainly oxygen gas(21% by volume) and water vapor (~ 1%).
- Liquid type: contain mainly water and dissolved oxygen.

Oxide layers

Protective	Non Protective
Non active	active
Adherent	Non adherent
Non porous, dense and compact	Porous
Ex: Al	Ex: iron rust

Factors affecting atmospheric corrosion:

- Type of metal or alloy
- Type of atmosphere (dust, humidity, pollutant. Water vapor, gases,…)
- Temperature (the corrosion rate increases with increasing temp.).
- Time (the extent of corrosion increases with increased exposure time).
- Surface Condition:
  - more roughness, more corrosion
  - The presence of foreign matter influence the speed of corrosion
  - The formation of a surface oxide layer and its properties

Uniform Corrosion:

Galvanic Corrosion

Causes:

- **Dissimilar metals**
- **Electrolyte (differential aeration corrosion)**

Mechanism of galvanic corrosion:

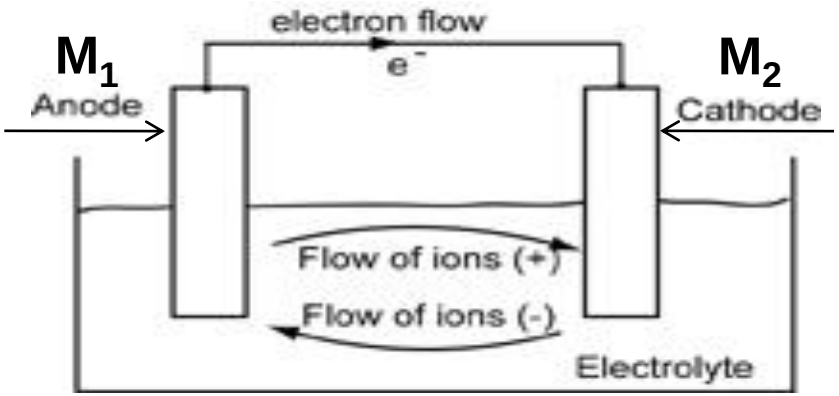
For two different metals M_1 and M_2 when exposed to the same corrosive environment, if M_1 is more active than M_2

- **Anodic reaction:**

The more active metal act as anode

Anodic metal will be corroded.

$M_1 \rightarrow M_1^{n+} + ne^-$ (n=1, 2, 3, …)



- **Cathodic reactions:**

(I) $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ $E^o = 1.23$ V O_2 rich acidic media

OR

(II) $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ $E^o = 0.401$ V O_2 rich non acidic media

OR

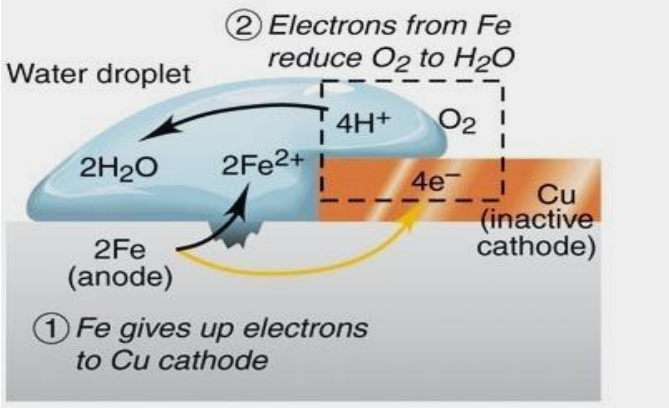
(III) $2H^+ + 2e^- \rightarrow H_2$ $E^o = 0.0$ V O_2 poor acidic media

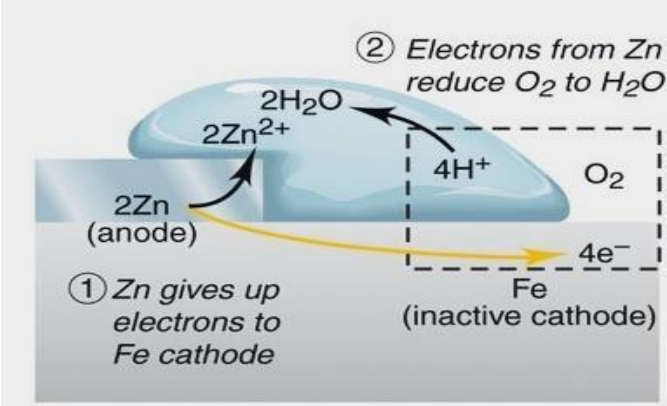
OR

(IV) $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ $E^o = -0.828$ V O_2 poor non acidic media

Examples of galvanic corrosion

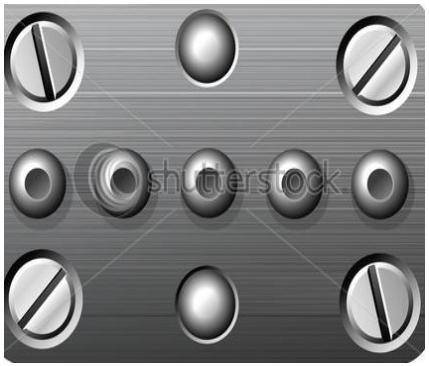
- **Metal plating**

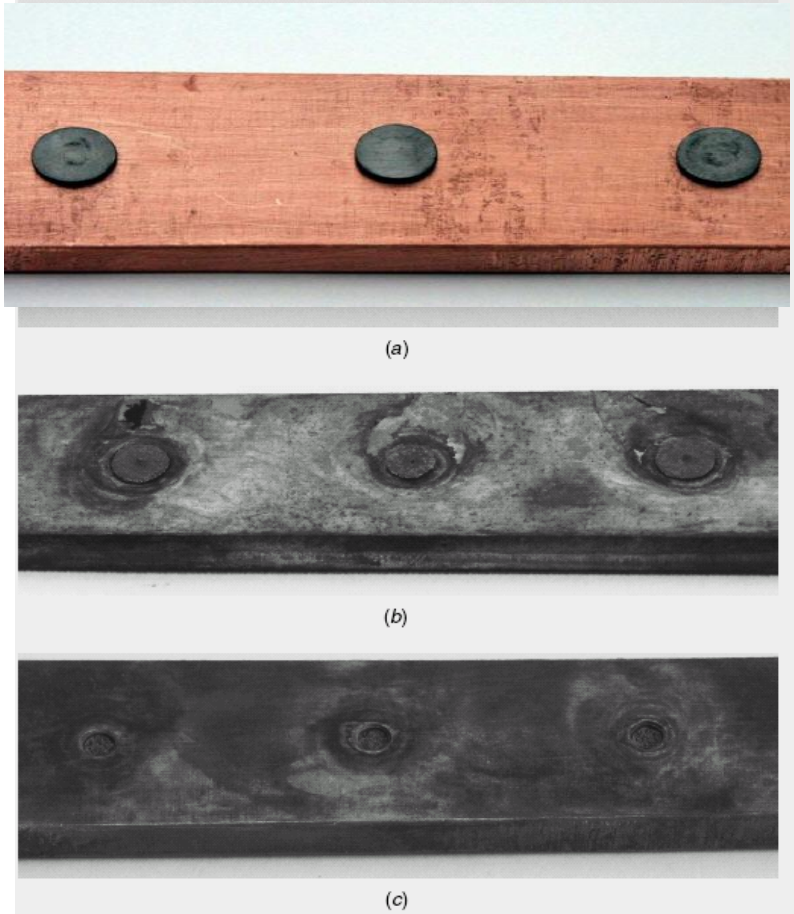




- **Steel Welding**

- **Steel rivet**



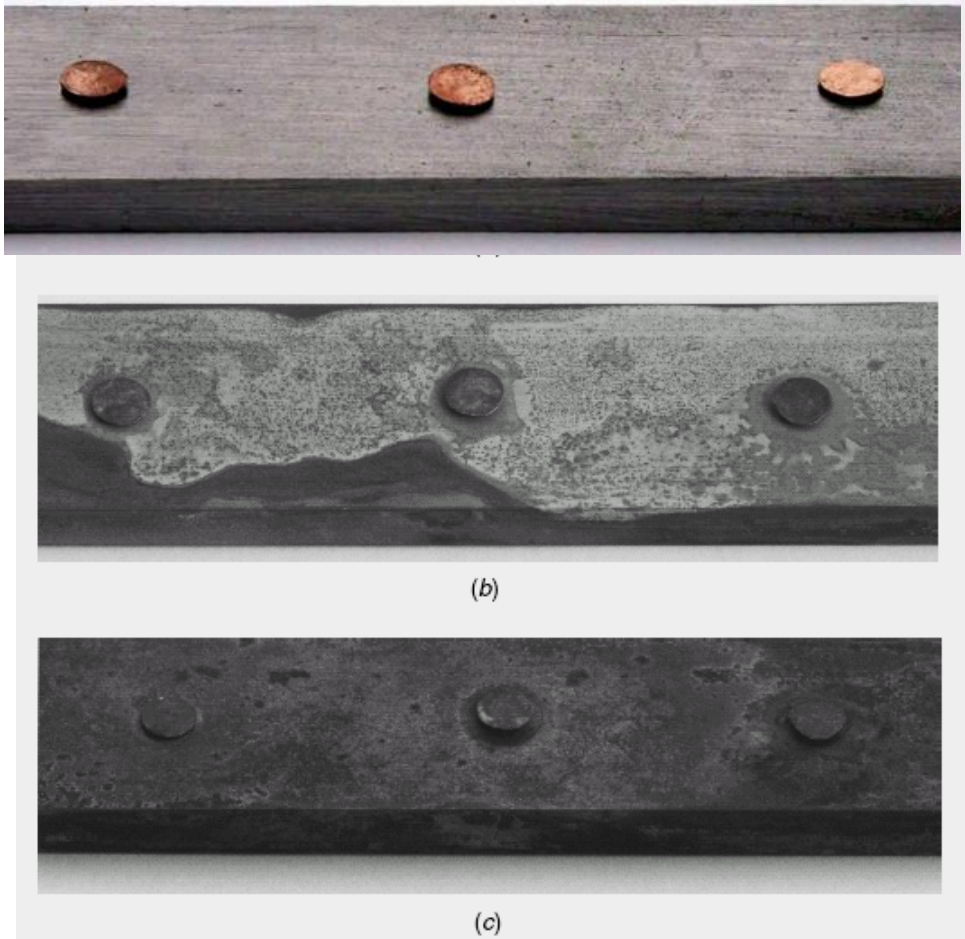


(a)

(b)

(c)

Fig. 2.14. Steel rivets on a copper bar: (*a*) at the start of the experiment; (*b*) 6 months after being submerged in 3% sodium chloride solution; and (*c*) after 10 months in the same solution.



(b)

(c)

Fig. 2.15. Copper rivets on a steel bar: (*a*) at the start of the experiment; (*b*) 6 months after being submerged in 3% sodium chloride solution; and (*c*) after 10 months in the same solution.

After immersion in Sea water for 6-10 months, the steel rivets are severely attacked & Cu-bar remains unchanged.

Large cathode/small anode

After immersion in Sea water for 6-10 months, the steel bar is severely attacked & Cu-rivets remain unchanged.

Small cathode/large anode

Small anode + large cathode = corrosion catastrophe

Factors affecting galvanic corrosion

Potential Difference (P.D.)

As the P.D increases the galvanic corrosion increase

The rate of corrosion of (Zn/Fe) galvanic cell (P.D.= 0.32V) is slower than corrosion of (Zn/Cu) galvanic cell (P.D.= 1.1V).

Distance

As distance increase as corrosion rate decrease.

Area

As the Area (C/A) ratio increases the galvanic corrosion increase.

As the Area (C/A) ratio decreases the galvanic corrosion decrease.

Small anode + large cathode = corrosion catastrophe

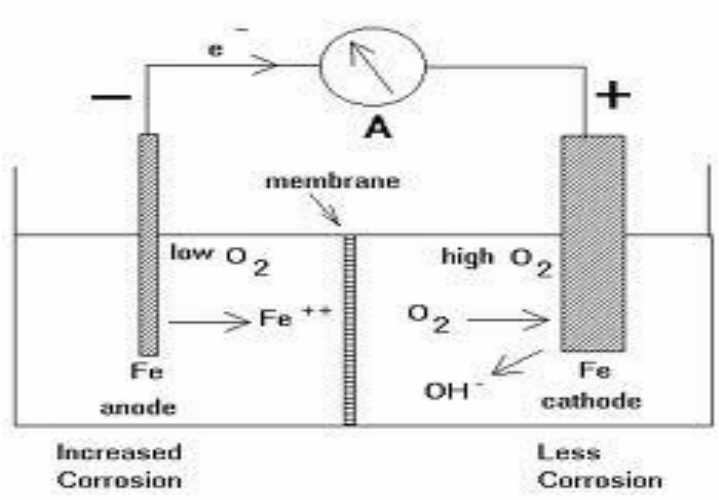
Uniform Corrosion:

Galvanic Corrosion

Differential Aeration Cell *(oxygenation)cell*

This type of corrosion takes place due to the difference in oxygen concentration on the same metal or alloy.

Differential aeration cells are concentration cells in which differing concentrations of oxygen is the driving force for corrosion. A differential aeration cell is shown in the Figure.

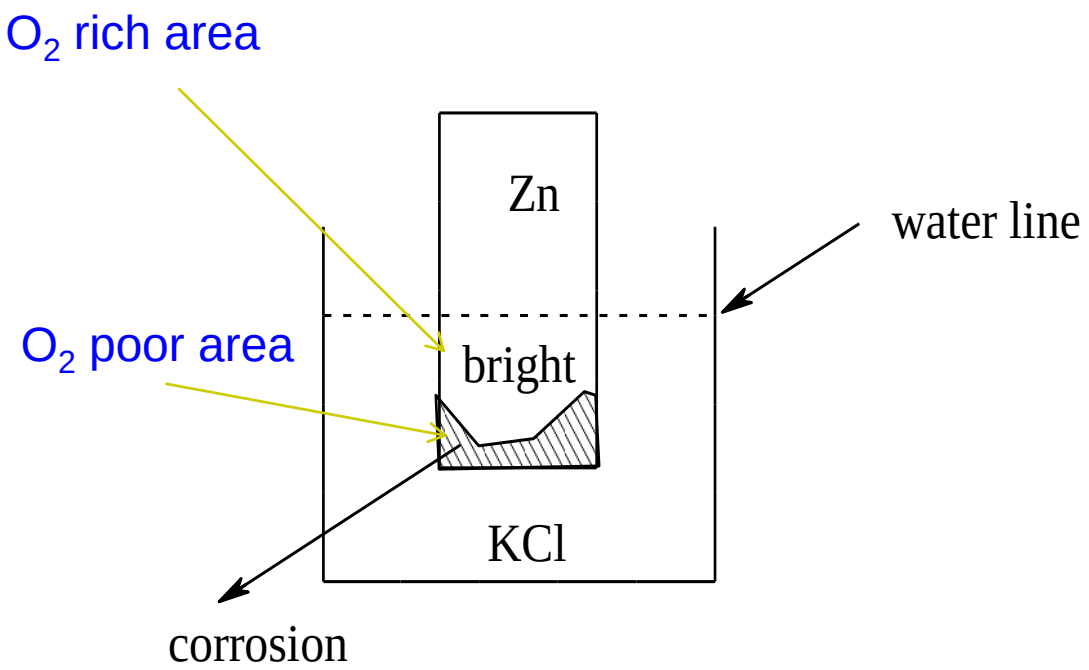


M | poor O_2 electrolyte || electrolyte rich O_2 | M

Anode (-) Corroded Area

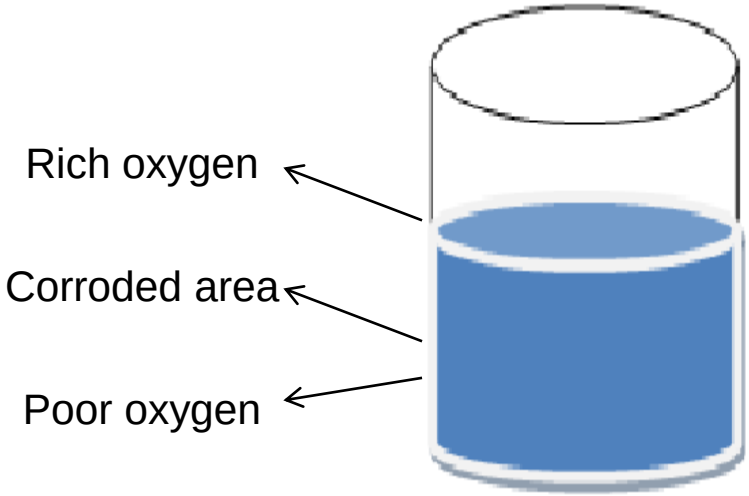
Cathode (+)

- *In a differential aeration cell, the electrode in lower O_2 pressure acts as the anode and the one in higher O_2 pressure acts as the cathode*

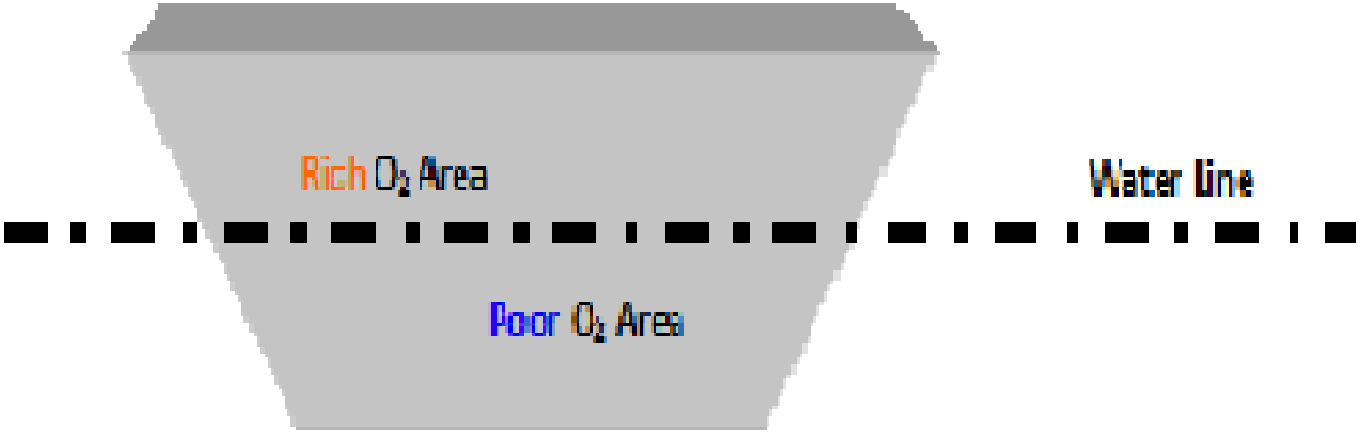


Partially immersed metallic plates.

Corrosion of water tanks



ship hull هيكل السفينة



Localized corrosion:
Crevice Corrosion

Causes: concentration gradients in electrolyte cause some areas high in ion concentrations that accelerate oxidation

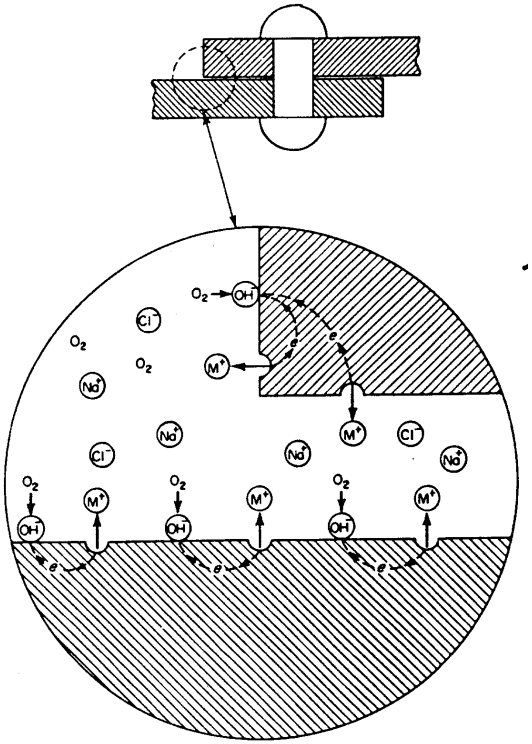
Crevice corrosion takes place in the following stages:

Initial stage:

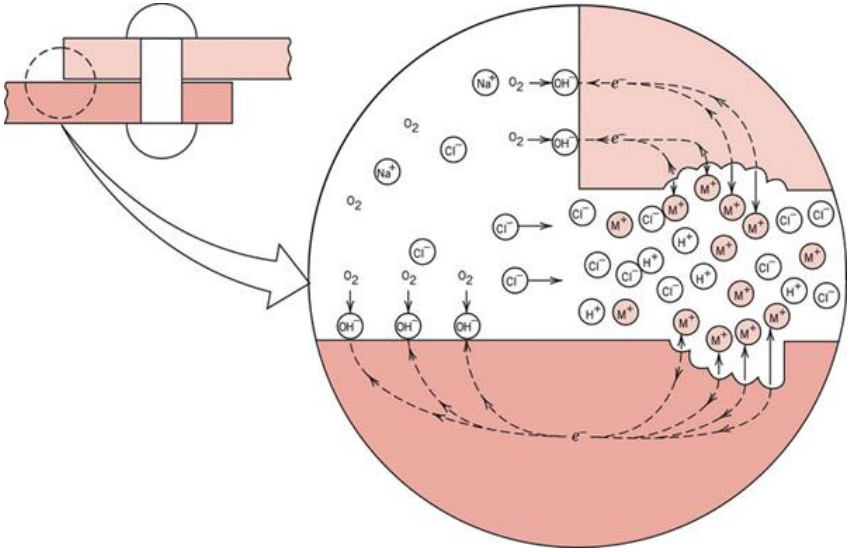
$M \rightarrow M^{z+} + ze^-$ anodic

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ cathodic

These reactions occur uniformly inside and outside the crevice as shown in the figure

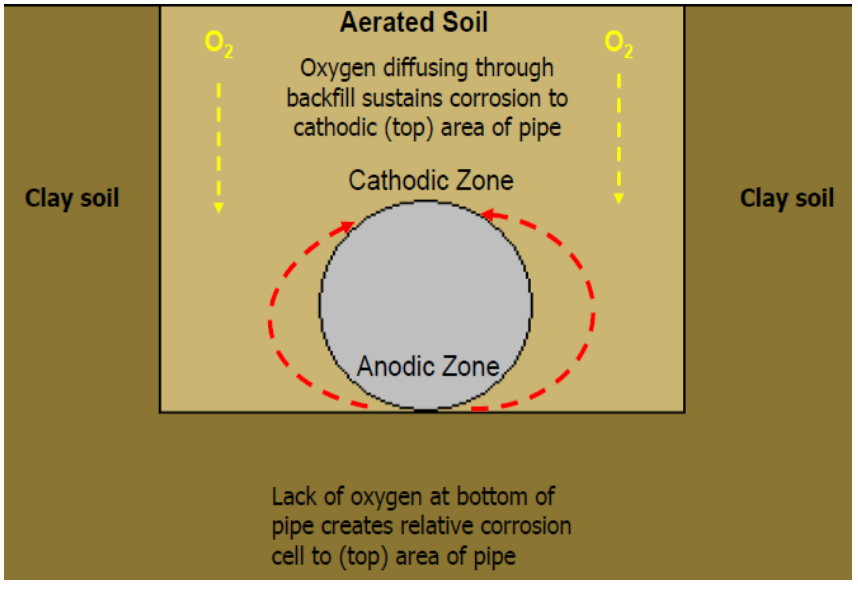


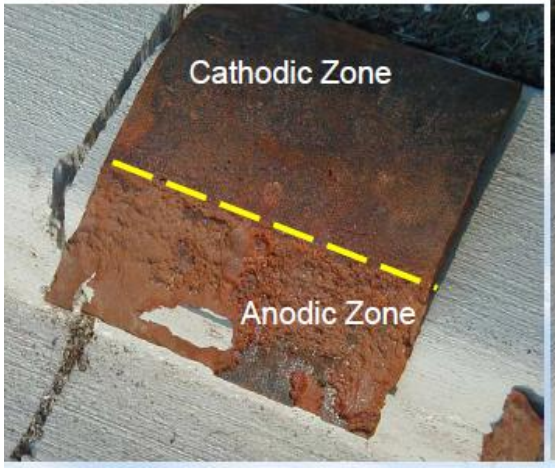
Later stage



Examples of Crevice or Concentration Cell

- Good example – crevices or under deposits of dirt or corrosion products where the solution is stagnant.
  - Crevice must be wide enough to allow solution to penetrate yet narrow enough for stagnancy (i.e. few thousandths of an inch).





Crevice corrosion between pipe and I-beam:

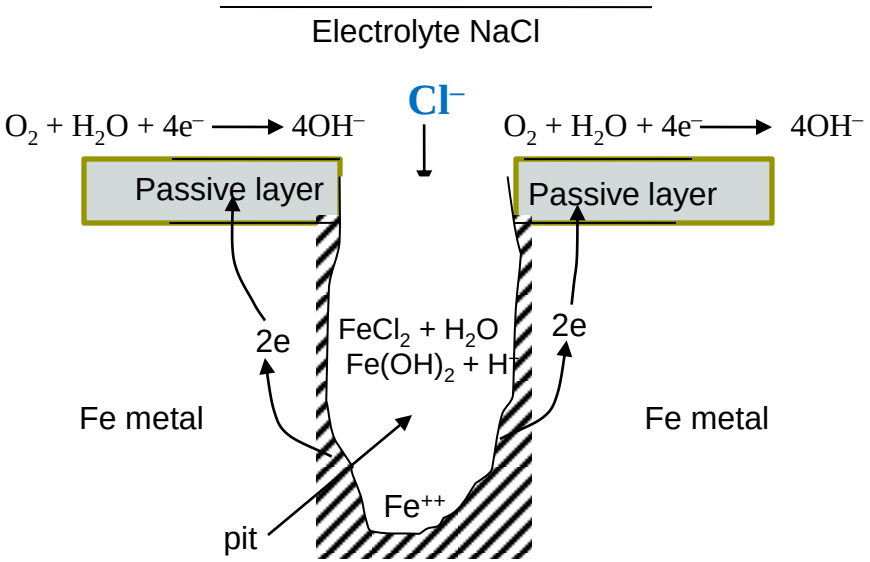
Rubber pads just accelerated the attack – why???

Localized corrosion:

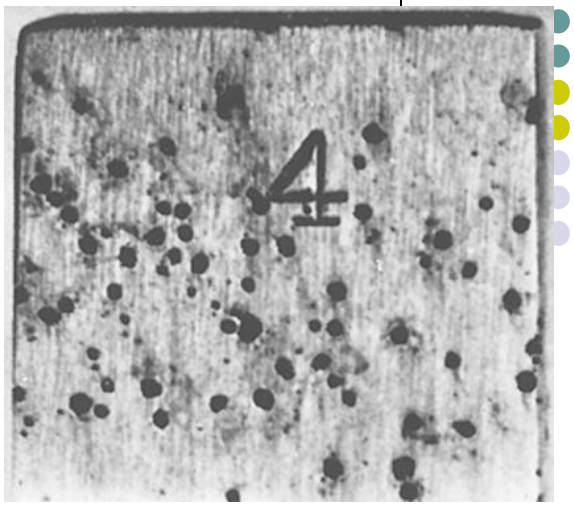
Pitting Corrosion

- Extremely localized corrosion that leads to the creation of small holes (pits) in the metal surfaces where, electrolyte is aggressive and stagnant.
- Pitting results from a local breakdown of the passive layer by aggressive anions.
- A galvanic cell is produced where the metal inside the pit acts as anode while the passive film acts as cathode as shown.
- $M^{z+} + zH_2O \rightarrow M(OH)_z + zH^+$

increases the corrosivity of the medium inside the pits

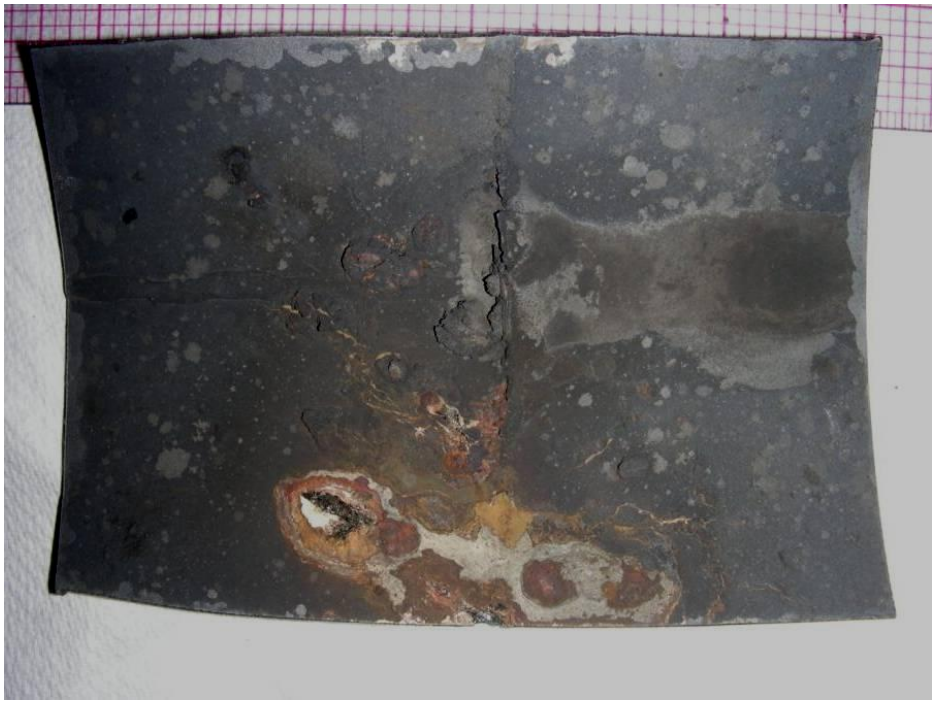


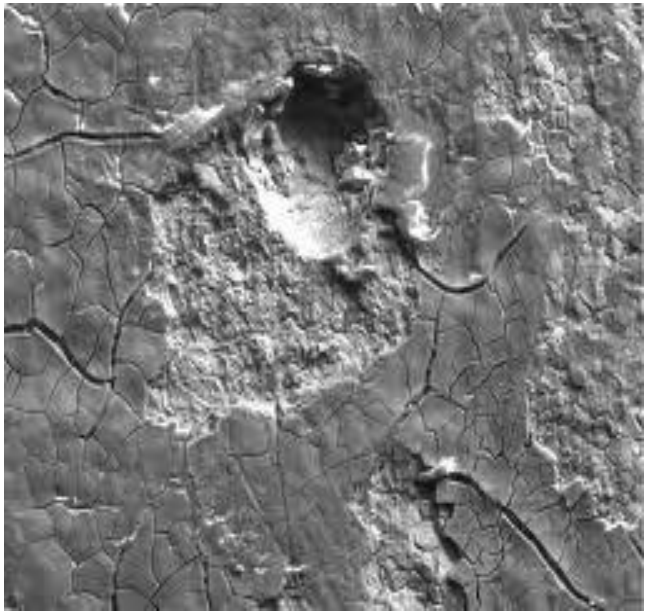
- Aggressive anions: Halides X^- ,SCN^-, SO_4^{2-} and ClO_4^-.
- Cl^- is the most important anion because of its wide spread distribution in natural water.
- Draws in chlorides and sulfates
- More of a problem in stagnant solutions.



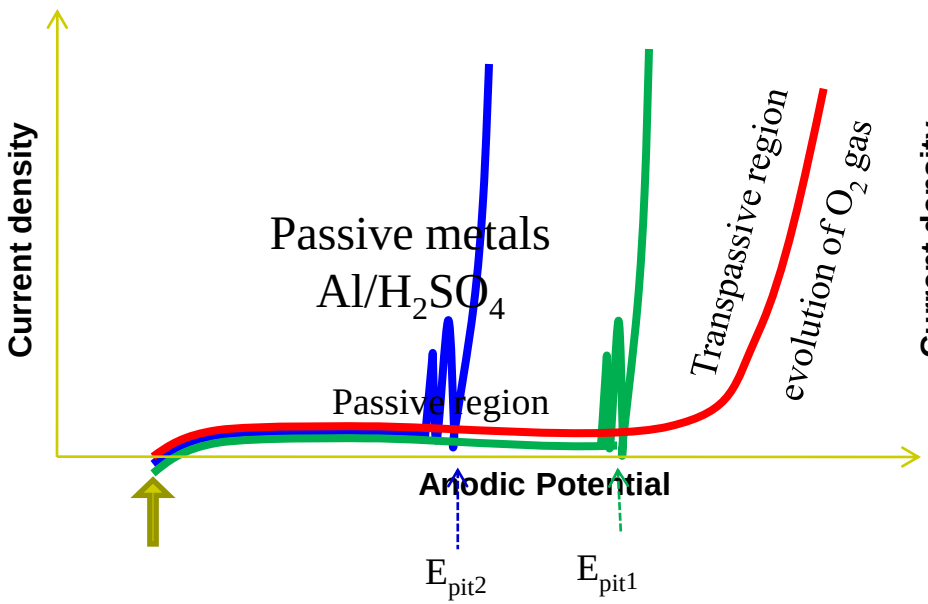
- Very destructive since highly localized – may go undetected until failure occurs.

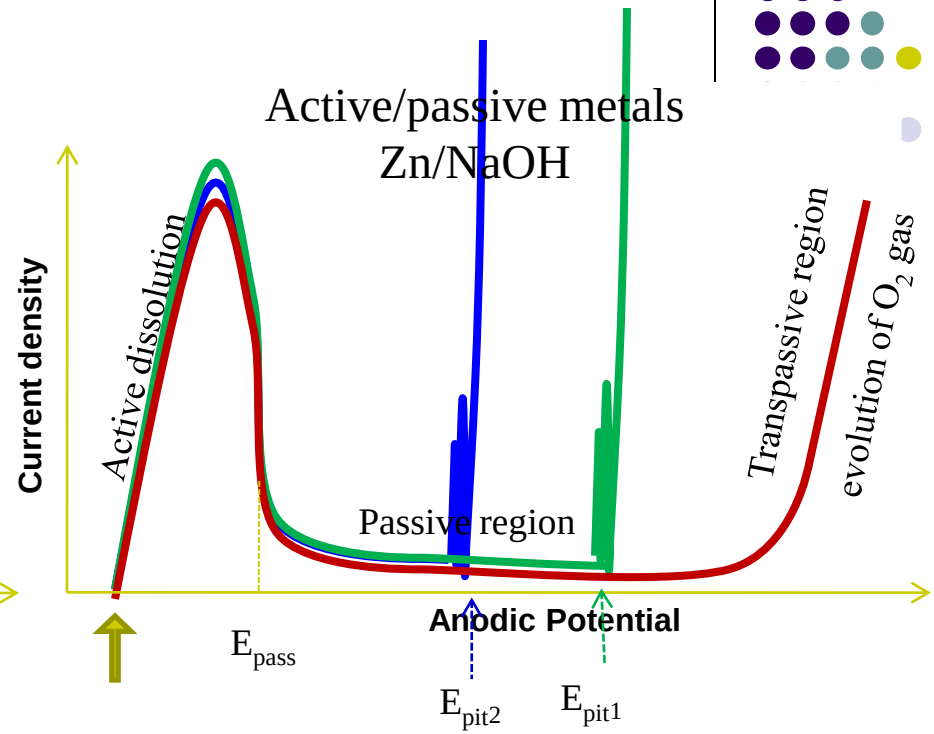
- Gravity causes pit to grow downward – corrosion rate can increase with time





Pitting potential, E_{pit}:
Potentiodynamic polarization





Cl^- aggressive anion causes pitting corrosion

At pitting potential (E_{pit}), the current rises suddenly indicates passivity breakdown and initiation of pitting corrosion

E_{pit} is more negative than potential corresponding to O_2 evolution reaction.

The pitting potential depends upon:

1. **Concentration and nature of the aggressive anion**
2. **Concentration of the electrolyte**
3. **pH of the medium**
4. **Temperature**

Any factor enhances pitting corrosion, shifts E_{pit} to more negative values

Localized corrosion:

Stress Corrosion Cracking SCC

Stressed material
(residual stress)
or external force

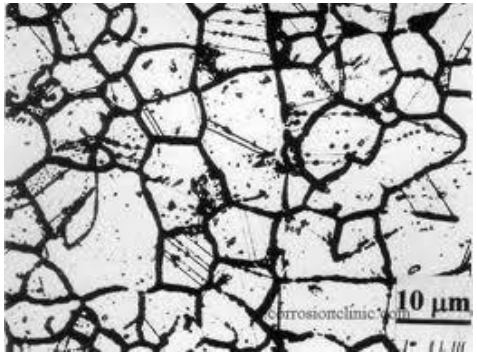
CORROSIVE

ENVIRONMENT

Cracking along interagranular path

The solution enters these cracks and causes SCC

Stress = force / cross sectional area



Intergranular

SCC

Examples

- brasses are sensitive to ammonia
- Iron or steels are sensitive to Hot, conc. alkali or NO^-_3 solution.
- Stainless steel are sensitive to Cl^-

Factors affecting SCC:

- Specific environment
- Stress magnitude and direction.
- Structure and metal composition.

Localized corrosion:

Erosion Corrosion

- **Erosion – corrosion is a process which involves conjoint erosion and corrosion.**
- **It is a combination between corrosion and mechanical wear on the metal surface.**
- **Erosion take place due to mechanical wear on metal surface**
- **Corrosion takes place due to chemical or electrochemical reactions.**
- **Caused by a rapid streaming liquid and the attack is thus dependent on the degree of turbulence.**
- **Erosion – corrosion can occur with most metals, but copper is particularly sensitive.**
- **In many cases the attack is due to the destruction of the protective layers or coating.**

Factors affecting erosion- corrosion

- **Metal surface (hardness of surface)**

Soft metals are more sensitive to erosion-corrosion.

Soft metal	Hard metal
Pb, Cu, Ag	Cr, high carbon steel

- **Environmental condition.**
  - **High degree of turbulent:**

Turbulent is formed due to high velocity of fluid and sudden change in direction of the fluid.

As turbulent increase as mechanical wear increase and hence the erosion will increase.

In order to decrease the turbulence effect , the thickness of tube or pipe in the turbulence area must be increased.

  - **Presence of solid or suspended materials:**

As suspended or solid material increase as the mechanical wear increase and hence erosion increase